ANALYSIS PROCESS

ASK - Define meaningful questions that guide the analysis, focusing on business goals and desired outcomes. This phase ensures that the analysis is aligned with stakeholder needs and generates relevant and actionable insights.

PREPARE- Collect relevant data from reliable sources and ensure there is sufficient information for analysis. Remove unnecessary or irrelevant data to maintain focus on key variables and improve efficiency.

PROCESS- Clean and prepare the data by handling missing values, removing outliers, and ensuring consistency. This step guarantees data accuracy and quality, enabling reliable analysis aligned with business objectives.

ANALYSIS- Analyze the data to identify patterns, trends, and relationships. Use SQL and analytical techniques to generate insights that answer key business questions and support decision-making.

SHARE- Present findings in a clear and structured way using visualizations and summaries. Communicate insights effectively to stakeholders to ensure understanding and facilitate data-driven decisions.

ACT- Use insights to make informed decisions and implement strategies. Monitor outcomes and adjust actions as needed to improve performance and achieve business goals.

1.- ASK

Superstore Sales Performance Analysis

Project Overview

This project analyzes historical retail transaction data to identify sales patterns, customer behavior, product performance, and revenue opportunities. The analysis aims to support strategic decision-making through data-driven insights.

Business Objective: Identify key factors affecting sales performance and uncover opportunities to increase revenue, improve profitability, and optimize business strategy.

1. General Analysis.
- What is total revenue generated?
- How many total orders were placed?
- What is overall sales performance?

2. Product Analysis.
- What are the top-performing products by revenue?
- Which products have the highest order frequency?
- Which products underperform?

3. Regional Analysis.
- Which regions generate the highest and lowest sales?
- Are there differences in purchasing patterns by region?

4. Customer Analysis.
- Who are the top customers by revenue?
- Is revenue concentrated among a small percentage of customers?
- What is customer purchasing behavior?

5. Trend Analysis.
- Are there seasonal sales trends?
- Which months perform best and worst?

6. Key Metrics.
- What is the average order value?
- How many unique customers exist?

Stakeholders

- Sales Managers.
- Marketing Team.
- Business Executives.

- Operations Leadership.

These stakeholders rely on performance insights to support strategic planning and commercial decisions.

Success Criteria

The analysis should provide clear, accurate, and actionable insights that help stakeholders improve sales performance, optimize strategies, and make data-driven decisions.

2.- PREPARE

In this phase, we collect and review the dataset to ensure it is suitable for analysis. The focus is on understanding the data structure, identifying relevant variables, and assessing its quality before proceeding to the data cleaning process.

Data Source

The dataset was obtained from Kaggle (Superstore Sales Dataset), a publicly available dataset commonly used for data analysis practice.

Dataset Description

The dataset contains sales transaction data, including variables such as order date, product name, category, region, customer information and sales.

Data Relevance

This dataset is relevant because it provides key information to analyze sales performance, identify trends, and evaluate different products and regions.

Data Limitations

Potential limitations include missing values, possible inconsistencies in data entry, and the lack of real-time updates.

3.- PROCESS

In this phase, the dataset is cleaned and prepared to ensure accuracy, consistency, and reliability. This includes handling missing values, checking for duplicates, validating data types, and identifying any inconsistencies or outliers that could affect the analysis.

Nulls Check

A data quality check was performed to identify missing values across the dataset. It was found that only the "Postal Code" column contained a small number of null values (11 out of 9800 rows). Due to their minimal impact, these were considered negligible for the analysis.

DUuplicate Check

A duplicate check was performed to ensure data quality and consistency. The dataset was analyzed by grouping all columns to identify any rows that were completely identical across every field.

This was achieved using a grouping method that compares all columns simultaneously and counts how many times each row appears. Only rows with more than one occurrence were considered duplicates.

No duplicate rows were found in the dataset, indicating that the data is clean and does not contain repeated records that could negatively impact the analysis.

It is important to note that repeated Order IDs were expected and do not represent duplicates, as each order may contain multiple products, resulting in multiple rows per transaction.

Data Consistency Check

Consistency checks were performed on key categorical variables such as Region, Category, Sub-Category, Segment, and Ship Mode. All values were found to be standardized and free of inconsistencies.

Date Validation

A validation of date fields was performed to ensure logical consistency and realistic timeframes. Order dates ranged from 2015 to 2018, while ship dates extended slightly beyond, which is expected due to order fulfillment timing. No cases were found where shipment occurred before the order date. Additionally, shipping times ranged from 0 to 7 days, with an average of approximately 4 days, indicating a consistent and reliable delivery process.

Outlier Analysis

An outlier analysis was conducted on the Sales variable by examining both high and low transaction values. The highest sales correspond to large product purchases, while the lowest values represent small or low-cost items. A distribution analysis showed that most transactions fall within low and medium ranges, with fewer high-value transactions. No outliers were identified as data errors, as all values reflect realistic business scenarios.

Final Data Quality Statement

Overall, the dataset was thoroughly cleaned and validated to ensure accuracy, consistency, and reliability. Missing values were minimal and did not significantly impact the analysis, no duplicate records were found, and all categorical variables were consistent and properly formatted. Numerical and date fields were also validated, showing logical ranges and no anomalies. Therefore, the dataset is considered reliable and ready for the analysis phase.

4.- ANALYSIS

Analysis Stage 1: General Sales Performance

Analytical Objective

The purpose of this analysis is to evaluate the overall sales performance of the business by measuring total revenue and establishing a baseline understanding of commercial activity. This provides a foundation for identifying trends, customer behavior, and future optimization opportunities.

Business Questions

This analysis answers the following questions:

- What is the total revenue generated?
- How many total orders were placed?
- How many unique customers exist?
- What is the average order value?

Business Question: **What is the total revenue generated?**

Interpretation

The dataset generated a total revenue of **$2,261,536.78**, indicating substantial commercial activity across the analyzed period. This reflects consistent transaction volume and suggests a broad customer purchasing base contributing to overall business performance.

Initial Insight

The total revenue indicates strong business activity and suggests that the company maintains a healthy level of market demand. However, total revenue alone does not reveal whether sales performance is evenly distributed across products, customers, or regions, requiring deeper segmentation analysis.

Business Relevance

Understanding total revenue establishes a baseline for evaluating overall business performance and supports further analysis aimed at identifying growth opportunities, high-performing segments, and areas requiring strategic improvement.

Business Question: **How many total orders were placed?**

Interpretation

The dataset contains 4,922 unique customer orders, representing the total number of completed transactions during the analyzed period. This indicates consistent sales activity across the business.

Initial Insight

The total number of orders reflects strong transaction volume and suggests active customer purchasing behavior. When compared with total revenue, this metric helps evaluate average order value and purchasing efficiency.

Business Relevance

Understanding total order volume provides context for measuring customer demand and supports further analysis of purchasing patterns and overall business performance.

Business Question: **What is the average order value?**

Interpretation

The average order value is $459.48, meaning that each customer order generates approximately this amount in revenue on average.

Initial Insight

This suggests that customer transactions are relatively high in value, indicating strong purchasing activity across the business. This metric provides visibility into transaction size and overall sales efficiency.

Business Relevance

Understanding average order value helps evaluate purchasing behavior and supports future strategies focused on increasing transaction value through product bundling, upselling opportunities, and targeted promotional efforts.

Stage 1 Summary Insight

The business demonstrates strong commercial performance, generating over **$2.26 million** in revenue across **4,922 orders** from **793 unique customers**, with an average order value of **$459.48**.

These metrics indicate consistent transaction activity supported by repeat customer purchasing behavior, establishing a strong foundation for deeper analysis of customer segmentation, product performance, and sales trends.

Analysis Stage 2: Product Performance Analysis

Analytical Objective

Evaluate product-level sales performance by identifying high-performing and underperforming products to uncover revenue-driving opportunities and product portfolio optimization strategies.

Business Questions

This analysis answers the following questions:

- What are the top-performing products by revenue?
- Which products have the highest order frequency?
- Which products underperform?

Business Question: **What are the top-performing products by revenue?**

Interpretation

The top 10 revenue-generating products produced **$244,620.20**, representing only **10.82%** of total company revenue. While the highest-performing product generated **$61,599.82**, overall revenue contribution is distributed across a wide range of products.

Initial Insight

This indicates a highly diversified revenue structure rather than heavy dependence on a small subset of products. Although premium office technology products lead individual rankings, overall business performance appears supported by broad product portfolio participation.

Business Relevance

A diversified revenue structure reduces operational risk and improves business stability. Strategic focus should prioritize maintaining balanced product performance across the catalog rather than relying exclusively on a few top-selling items.

Business Question: **Which products have the highest order frequency?**

Interpretation

The products with the highest order frequency are primarily low-cost office consumables such as staples, envelopes, paper, and binders. These products differ significantly from the highest-revenue products, which are dominated by premium office equipment and technology solutions.

Initial Insight

This indicates a dual product-performance structure within the business. Frequently purchased consumables drive consistent transactional volume, while high-value equipment generates substantial revenue through lower-frequency, high-ticket purchases.

Business Relevance

TThe business should apply different strategies depending on product behavior. Frequently purchased consumables should remain consistently available to encourage repeat purchases, while high-value equipment sales require stronger customer relationships and personalized sales efforts to maximize large transactions.

Business Question: Which products underperform?

Interpretation

The lowest-performing products generated extremely low total revenue, ranging from approximately **$1.62 to $8.10**. These products are primarily low-cost office consumables such as envelopes, paper products, highlighters, and small desk accessories.

Initial Insight

The minimal revenue contribution suggests these products either experience limited purchasing frequency, carry very low unit prices, or occupy niche roles within the broader product catalog. Their financial impact on overall business performance appears negligible.

Business Relevance

Products with minimal revenue contribution should be evaluated to determine their strategic role within the portfolio. The business may consider whether these items support complementary purchasing behavior, require stronger visibility, or represent candidates for reduced operational focus.

Stage 2 Summary Insight

The product performance analysis revealed a diversified revenue structure across the product portfolio, with the top 10 products contributing only **10.82%** of total revenue. High-revenue products are primarily premium office technology solutions, while the most frequently purchased items consist of lower-cost consumable office supplies.

This distinction indicates a dual product-performance model in which high-ticket products drive substantial revenue through lower transaction frequency, while consumable products sustain recurring transactional volume. Additionally, underperforming products contribute minimally to total sales and should be strategically evaluated to determine their operational value within the broader portfolio.

Overall, the findings suggest that maintaining product portfolio balance is essential for sustaining revenue stability, optimizing inventory strategy, and supporting differentiated commercial initiatives across both premium and recurring-demand product segments.

Analysis Stage 3: Regional Analysis

Analytical Objective

Evaluate regional sales performance to identify geographic revenue distribution patterns and uncover regional differences that may influence strategic resource allocation, market expansion opportunities, and localized sales optimization efforts.

Business Questions

This analysis answers the following questions:

- Which regions generate the highest and lowest sales?
- Are there significant differences in purchasing behavior across regions?

Business Question: Which regions generate the highest and lowest sales?

Interpretation

Regional sales performance shows moderate variation across all four regions. The West and East regions lead total revenue generation, while Central and South contribute lower but still substantial portions of overall business sales.

Initial Insight

This suggests a relatively balanced geographic revenue distribution with stronger commercial performance concentrated in the West and East markets. The absence of extreme regional concentration indicates diversified market reach and reduced dependency on a single geographic segment.

Business Relevance

Understanding regional performance differences helps guide resource allocation, localized sales strategy development, and market expansion planning. Lower-performing regions may present opportunities for targeted growth initiatives, while stronger regions can provide benchmarks for replicating successful commercial practices.

Business Question: Are there significant differences in purchasing behavior across regions?

Interpretation

Regional performance differs significantly across transaction behavior. The West region leads total revenue through the highest order frequency, while East and South generate larger average transaction values.

Initial Insight

TThis indicates distinct regional purchasing dynamics. West demonstrates stronger transactional activity and customer volume, while East and South reflect fewer but higher-value purchasing behavior.

Business Relevance

Recognizing regional behavioral differences enables more precise strategic planning. High-volume regions may benefit from customer retention and operational scalability initiatives, while high-value regions present opportunities for premium product positioning and targeted high-ticket sales strategies.

Stage 3 Summary Insight

The regional sales analysis revealed a relatively balanced geographic revenue distribution across all four business regions, with the **West region generating the**

highest total revenue and the **South region contributing the lowest overall sales performance.**

Further analysis showed that regional performance differences are driven by distinct purchasing behaviors rather than simple revenue dominance. The West region leads through **higher transaction volume**, while East and South demonstrate **higher average order values despite lower order frequency.**

These findings suggest that regional sales performance is influenced by varying customer purchasing dynamics, requiring differentiated commercial strategies. High-volume regions may benefit from operational scalability and retention-focused initiatives, while higher-value regions present opportunities for premium product positioning and targeted high-ticket sales optimization.

Analysis Stage 4: Customer Analysis

Analytical Objective

Evaluate customer purchasing behavior to identify high-value customers, measure revenue concentration patterns, and uncover customer-level purchasing dynamics that support retention strategy, customer segmentation, and long-term revenue optimization.

Business Questions

This analysis answers the following questions:

- Who are the top customers by revenue?
- Is revenue concentrated among a small percentage of customers?
- What is customer purchasing behavior?

Business Question: **Who are the top customers by revenue?**

Interpretation

The highest-revenue customer is **Sean Miller**, generating **$25,043.05**, followed by **Tamara Chand ($19,052.22)** and **Raymond Buch ($15,117.34)**. The top 10 customers each generated between **$12,129 and $25,043** in total revenue. Revenue among top customers shows moderate variation without extreme concentration at the individual customer level. The revenue gap between the highest and lowest top-10 customers remains relatively controlled.

Initial Insight

This suggests a diversified high-value customer base rather than dependence on a single dominant account. Revenue generation appears supported by multiple strong customer relationships across the customer portfolio.

Business Relevance

Identifying top customers supports retention strategy development and relationship prioritization. Maintaining engagement with these high-value accounts can strengthen revenue stability while reducing the risk associated with customer churn among strategically important clients.

Business Question: **Is revenue concentrated among a small percentage of customers?**

Interpretation

The top 10 highest-revenue customers generated **$153,811.17**, representing only **6.8%** of total business revenue. Revenue concentration among top customers is relatively low. Even the highest-value customer accounts collectively contribute only a small portion of total overall revenue.

Initial Insight

This indicates a highly diversified customer revenue structure with limited dependency on individual accounts. Business performance appears broadly distributed across the customer base rather than concentrated among a small number of dominant customers.

Business Relevance

A diversified customer portfolio reduces revenue concentration risk and strengthens long-term business resilience. This structure minimizes financial exposure to customer churn while supporting more stable and predictable revenue generation across the broader customer base.

Business Question: **What is customer purchasing behavior?**

Interpretation

Customers placed an average of **6.21 orders during the four-year period (2015–2018)**, with the highest concentration of customers placing between **4 and 7 orders**. Customer purchasing frequency is relatively consistent across the customer base, with most customers demonstrating repeat transactional behavior rather than isolated one-time purchases.

Initial Insight

Customer purchasing behavior reveals multiple high-value customer profiles, including strategic high-ticket buyers, highly recurrent moderate-spending customers, and balanced premium accounts. This suggests revenue stability is supported through diverse purchasing patterns rather than a single dominant customer behavior model. .

Business Relevance

Recognizing distinct customer value segments enables more targeted relationship management strategies. High-ticket customers may benefit from personalized account support, while recurrent customers present opportunities for retention-focused engagement and long-term value expansion initiatives.

Stage 4 Summary Insight

The customer analysis revealed a highly diversified revenue structure supported by a broad and stable customer base. The top 10 customers contributed only **6.8% of total business revenue**, indicating limited concentration risk and reducing dependency on individual high-value accounts.

Customer purchasing activity shows consistent repeat behavior across the portfolio, with an average of **6.21 orders per customer during the 2015–2018 period**. Most customers placed between **4 and 7 orders**, reflecting stable recurring engagement rather than isolated transactional behavior.

Further behavioral analysis identified distinct high-value customer segments, including **strategic high-ticket buyers**, **high-frequency recurring customers**, and **balanced premium accounts**. This diversity in purchasing patterns suggests that revenue stability is supported by multiple customer value profiles rather than a single dominant purchasing behavior.

Overall, the customer portfolio demonstrates strong commercial resilience and presents opportunities for segmented relationship management strategies focused on retention, personalized engagement, and long-term customer lifetime value optimization.

Analysis Stage 5: Trends analysis

Analytical Objective

Evaluate sales trends over time to identify seasonal patterns, performance fluctuations, and temporal opportunities that support forecasting and strategic sales planning.

Business Questions

This analysis answers the following questions:

- Are there seasonal sales trends?
- Which months perform best and worst?

Business Question: **Are there seasonal sales trends?**

Interpretation

The data shows a very clear and consistent seasonal pattern across all analyzed years. Sales tend to be lower at the beginning of the year, especially in January and February, and then gradually increase as the year progresses. This growth becomes more pronounced in the second half of the year, reaching its peak consistently in the last quarter, particularly between October and December. This behavior is repeated every year in the dataset, indicating that it is not random variation but a structural seasonal pattern within the business.

Initial Insight

The business exhibits a strong and predictable seasonal pattern where performance is heavily concentrated in the last quarter of the year. Sales progressively increase throughout the year, with a consistent acceleration in Q3 and a clear peak in Q4 across all years analyzed. This indicates that revenue is not evenly distributed but instead follows a stable annual cycle driven by seasonal demand.

Business Relevance

This seasonality is highly important for strategic planning. The company should prioritize inventory buildup and operational readiness before Q4 to avoid stockouts during peak demand. Marketing efforts should be intensified in Q3 to maximize conversion into the high-performing last quarter. Additionally, financial forecasting models should account for this strong seasonality to improve accuracy, while cash flow planning must consider weaker performance in Q1 to maintain stability throughout the year.

Business Question: **Which months perform best and worst?**

Interpretation

The aggregated monthly sales across all years show a clear ranking of performance. **November is the highest-performing month**, followed by **December and September**. On the opposite end, **February is the lowest-performing month**, followed by January and April. This indicates a strong concentration of revenue in the last quarter of the year, while the first months consistently underperform. The data reveals a very clear and stable pattern where the highest sales consistently occur in the final months of the year, particularly November and December, which dominate overall performance. September and October also show strong results, reinforcing the strength of Q4. In contrast, the beginning of the year shows significantly weaker performance, with February and January being the lowest months across the entire dataset. This confirms a strong and repeatable seasonal cycle in business activity.

Initial Insight

The business is heavily driven by end-of-year demand, with November acting as the peak sales month and December closely following. This suggests that customer purchasing behavior is strongly influenced by seasonal factors, likely including holidays, promotions, or year-end budgeting cycles. Meanwhile, early-year months represent a predictable slowdown period, which can be used strategically for planning rather than growth-focused campaigns.

Business Relevance

Understanding the best and worst performing months is critical for optimizing business strategy. The company should allocate maximum marketing budget and inventory capacity toward Q4, especially November and December, to fully capture peak demand. Operational teams should prepare for increased workload during this period to avoid bottlenecks. Conversely, Q1 should be used for efficiency improvements, cost control, and preparation for the annual growth cycle, rather than aggressive expansion efforts. This seasonal awareness enables better forecasting, resource allocation, and revenue optimization throughout the year.

Stage 5 Summary: Trend Analysis

Stage 5 analyzed sales trends over time to identify seasonal patterns and performance fluctuations across months and years.

The data shows a strong and consistent seasonal pattern. Sales gradually increase throughout the year, with the highest performance concentrated in the last quarter (Q4), especially in November and December. In contrast, the first quarter—particularly January and February—shows the lowest sales levels. This pattern repeats across all years, confirming a stable yearly cycle.

The business is highly seasonal, with Q4 being the main revenue driver and Q1 being the weakest period. This indicates predictable demand cycles that can be used for better planning and forecasting.

This seasonality should guide inventory planning, marketing strategy, and forecasting models. Resources should be focused on maximizing Q4 performance, while Q1 should be used for optimization and preparation.

Stage 6: Keys Metrics Analysis

Analytical Objective

Define and evaluate core business performance indicators, including average order value, customer base size, and purchase frequency, in order to understand revenue structure, customer behavior, and transactional efficiency. These metrics provide a foundational view of how value is generated and how frequently customers engage with the business.

Business Questions

This analysis answers the following questions:

- What is the average order value?

- How many unique customers exist?
- What is the average customer purchase frequency?

Business Question: What is the average order value?

Interpretation

The average order value indicates that, on average, each order generates approximately **$459 in sales**. This reflects a moderate transaction size, suggesting that customers are not making very small purchases but also not extremely high-value bulk orders.

Initial Insight

The business operates with a mid-range order value, meaning revenue growth is likely driven more by increasing order volume and customer retention rather than high-ticket purchases. Small improvements in order frequency or basket size could have a significant impact on total revenue.

Business Relevance

Understanding the average order value is essential for pricing strategy, marketing campaigns, and sales forecasting. Increasing this metric through bundling, upselling, or cross-selling could directly improve revenue without needing to acquire additional customers.

Business Question: How many unique customers exist?

Interpretation

The business serves a moderately sized customer base of 793 distinct customers. This indicates a fairly broad customer reach, suggesting that revenue is distributed across a relatively diverse group rather than being dependent on a small number of high-value clients.

Initial Insight

The customer base size suggests that growth opportunities may come more from increasing customer retention and repeat purchases rather than relying solely on acquisition. Since the customer pool is not extremely large, improving engagement with existing customers could significantly impact total revenue.

Business Relevance

Knowing the number of unique customers helps define the scale of the business and informs marketing strategy. It supports decisions around customer acquisition budgets, retention strategies, and segmentation for targeted campaigns.

Stage 6 Summary: Key Metrics

Overview

Stage 6 focused on evaluating the core business metrics to understand revenue generation, customer base size, and overall transactional structure of the business.

The average order value is **$459.48**, indicating that each transaction contributes a moderate-to-high revenue per order, suggesting that the business relies on meaningful order sizes rather than low-value purchases.

The dataset includes **793 unique customers**, showing a relatively broad customer base with a diverse distribution of sales across multiple clients rather than dependence on a small group of customers.

The business is driven by a combination of a solid customer base and a healthy average order value, meaning revenue is primarily generated through consistent order value rather than extreme customer concentration.

These metrics are essential for understanding the structure of revenue. The business should prioritize maintaining or increasing average order value through pricing strategies, bundling, or upselling, while also focusing on customer acquisition and retention to expand the existing customer base.

5.- SHARE

Executive Summary

This analysis evaluated sales performance between 2015 and 2018 using the Superstore Sales dataset. During the analyzed period, the business generated **$2,261,536.78** in total revenue across **4,922 orders** from **793 unique customers**, with an average order value of **$459.48**.

Product analysis revealed a diversified revenue structure, with the top 10 products contributing only **10.82%** of total revenue. High-revenue products were primarily premium office equipment, while the most frequently purchased products consisted mainly of low-cost office consumables.

Regional analysis showed that the **West region generated the highest revenue ($710,219.68)**, while the **South region generated the lowest ($389,151.46)**. Further analysis indicated that the West region achieved its revenue leadership through higher order volume rather than larger transaction sizes. In contrast, East and South customers exhibited higher average order values.

Customer analysis demonstrated a highly diversified customer base. The top 10 customers generated only **6.8%** of total revenue, indicating low customer concentration risk. Additionally, customers placed an average of **6.21 orders during the four-year period**, with most customers placing between **4 and 7 orders**, suggesting stable repeat purchasing behavior.

Overall, the business demonstrates strong revenue diversification across products, customers, and regions, supporting long-term commercial resilience and reducing dependency on individual revenue sources.

Key Findings

1. Revenue Performance

Finding

The business generated **$2,261,536.78** in total revenue from **4,922 orders** and **793 customers** during the analyzed period.

Business Impact

The company maintains substantial commercial activity supported by a broad

customer base and consistent transaction volume.

2. Product Performance

Finding

The top 10 products generated **10.82%** of total revenue, indicating that revenue is distributed across a wide product portfolio.

Business Impact

Revenue diversification reduces dependency on individual products and contributes to overall business stability.

3. Regional Performance

Finding

The West region generated the highest revenue while also recording the highest number of orders. However, East and South regions achieved higher average order values.

Business Impact

Regional performance is influenced by different purchasing behaviors, suggesting opportunities for region-specific sales and marketing strategies.

4. Customer Performance

Finding

The top 10 customers contributed only **6.8%** of total revenue, while most customers placed between **4 and 7 orders** during the analyzed period.

Business Impact

Low customer concentration risk strengthens revenue stability and reduces dependence on individual customer accounts.

5. Customer Purchasing Behavior

Finding

Customers placed an average of **6.21 orders** during the four-year period, demonstrating recurring purchasing activity across the customer base.

Business Impact

Repeat purchasing behavior supports long-term revenue generation and creates opportunities to increase customer lifetime value through retention initiatives.

Visualizations Included

The final dashboard includes the following visualizations:

- Total Revenue KPI
- Total Orders KPI
- Total Customers KPI
- Average Order Value KPI
- Revenue by Region
- Top Products by Revenue
- Top Customers by Revenue
- Monthly Sales Trend
- Customer Revenue Distribution

- Product Revenue Distribution

Stakeholder Takeaways

Sales Managers

Focus on retaining high-value customers and increasing purchase frequency among existing accounts to maximize customer lifetime value.

Marketing Team

Increase visibility of underperforming products while continuing to support high-performing product categories through targeted promotional efforts.

Business Executives

Leverage the business's diversified revenue structure and balanced regional performance to support sustainable long-term growth and reduce concentration risk.

6.- ACT

Strategic Recommendations

Recommendation 1: Strengthen Customer Retention

Develop customer retention initiatives focused on high-value and repeat customers to increase customer lifetime value and maintain revenue stability.

Recommendation 2: Replicate Successful Regional Practices

Analyze commercial strategies used in the West region and evaluate opportunities to apply successful practices to lower-performing regions.

Recommendation 3: Optimize Product Portfolio Management

Evaluate underperforming products to determine whether they support strategic business objectives, cross-selling opportunities, or require reduced operational focus.

Recommendation 4: Increase Purchase Frequency

Implement customer engagement programs, targeted promotions, and personalized marketing campaigns designed to encourage more frequent purchases.

Recommendation 5: Continue Monitoring Revenue Diversification

Maintain a balanced approach to customer, product, and regional management to preserve the diversified revenue structure identified throughout the analysis.

Next Steps

Future analysis should focus on:

- Customer segmentation using RFM analysis
- Product profitability analysis
- Customer lifetime value estimation
- Sales forecasting models
- Regional growth opportunity analysis
- Predictive analytics and machine learning applications